

The equation of a straight line [Cambridge revision]

Gradient, midpoint and length — three formulas that answer almost every coordinate question.

LESSON 53–54

2 × 40 MIN

REVISION OF GRADE 8–9 COORDINATES

P1 3.1–3.3

LESSON CLOCK

15'

25'

25'

20'

10'

The three formulas	15 min
Forms of the equation	25 min
Parallel and perpendicular	25 min
The perpendicular bisector	20 min
Homework	10 min

LEARNING OBJECTIVES

- Find the length, midpoint and gradient of a line segment.
- Write the equation of a line through a point with a given gradient.
- Use the conditions for parallel and perpendicular lines.
- Find the perpendicular bisector of a segment and the foot of a perpendicular.

TEXTBOOK

National	Geometry 9, pp. 96–118
Cambridge	Pure Mathematics 1, pp. 44–56
Workbook	Exercise 3A–3C

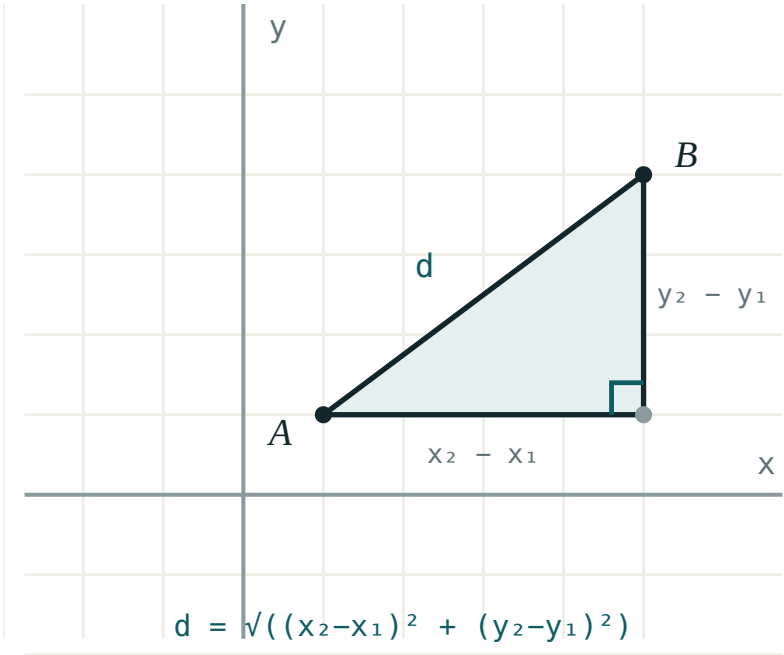
01

Explanation

The three formulas

For $A(x_1, y_1)$ and $B(x_2, y_2)$:

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) \quad m = \frac{y_2 - y_1}{x_2 - x_1}$$



The distance is the hypotenuse of a right triangle with
legs Δx and Δy .

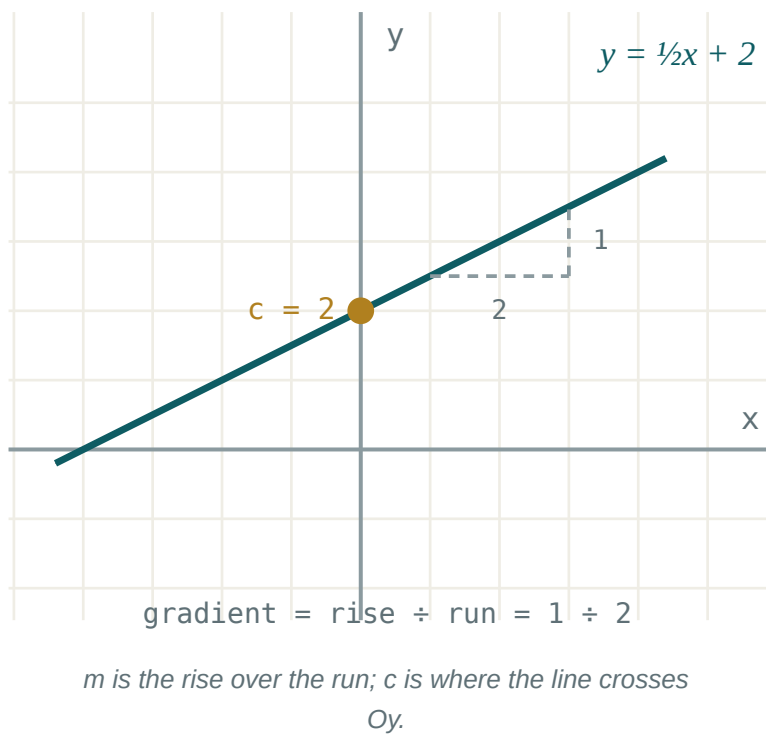
A	B	AB	M	M
(1, 2)	(4, 6)	5	(2.5, 4)	$\frac{4}{3}$
(-2, 3)	(4, -5)	10	(1, -1)	$-\frac{4}{3}$
(0, 0)	(6, 8)	10	(3, 4)	$\frac{4}{3}$

SUBTRACT IN THE SAME ORDER, TWICE

Gradient is $\frac{y_2 - y_1}{x_2 - x_1}$, not $\frac{y_2 - y_1}{x_1 - x_2}$. Reversing one subtraction and not the other flips the sign — and a positive gradient reported as negative loses every mark that follows.

Forms of the equation

FORM	EQUATION	USE IT WHEN
gradient–intercept	$y = mx + c$	the intercept is known or wanted
point–gradient	$y - y_1 = m(x - x_1)$	a point and a gradient are given
two points	$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$	two points are given
general	$ax + by + c = 0$	an answer must be tidy, or the line is vertical



THE VERTICAL LINE HAS NO GRADIENT

$x = 3$ cannot be written as $y = mx + c$ — its gradient is undefined, not infinite and not zero. A horizontal line, by contrast, has gradient 0 and the perfectly ordinary equation $y = 4$.

Parallel and perpendicular

parallel: $m_1 = m_2$ perpendicular: $m_1 m_2 = -1$

The second condition says the gradients are **negative reciprocals**: $\frac{2}{3}$ pairs with $-\frac{3}{2}$, and 4 with $-\frac{1}{4}$.

LINE	PARALLEL THROUGH $(1, 5)$	PERPENDICULAR THROUGH $(1, 5)$
$y = 2x + 1$	$y = 2x + 3$	$y = -\frac{1}{2}x + \frac{11}{2}$
$y = -\frac{3}{4}x$	$y = -\frac{3}{4}x + \frac{23}{4}$	$y = \frac{4}{3}x + \frac{11}{3}$
$3x + 2y = 7$	$3x + 2y = 13$	$2x - 3y + 13 = 0$

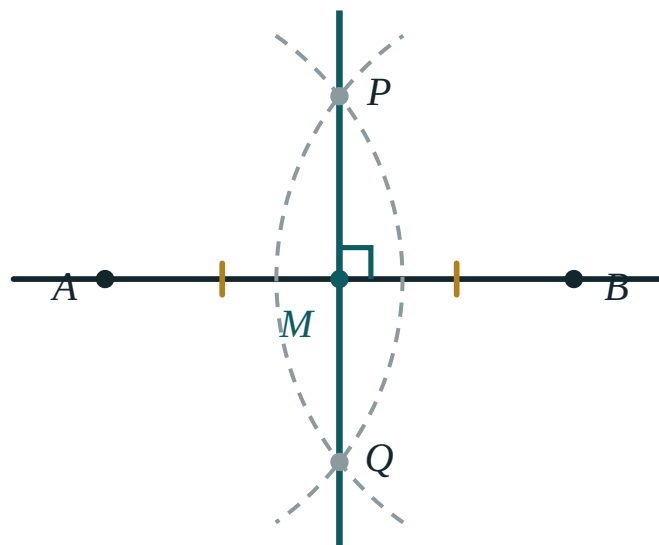
THE SHORTCUT FOR THE GENERAL FORM

A line parallel to $ax + by = c$ is $ax + by = k$; one perpendicular to it is $bx - ay = k$. Swap the coefficients and change one sign — no gradient needed at all.

The perpendicular bisector

The perpendicular bisector of AB is the set of points equidistant from A and B . Two steps build it:

1. the midpoint M of AB — it passes through this;
2. the negative reciprocal of the gradient of AB — this is its gradient.



Every point of the bisector is the same distance from A as from B .

Example. $A(1, 2)$, $B(5, 8)$. Then $M(3, 5)$ and $m_{AB} = \frac{3}{2}$, so the bisector has gradient $-\frac{2}{3}$:

$$y - 5 = -\frac{2}{3}(x - 3), \text{ that is } 2x + 3y = 21$$

TWO CONDITIONS, BOTH NEEDED

A line through M that is not perpendicular is not the bisector; a perpendicular line that misses M is not either. Check both before writing the answer.

02 Worked examples

EXAMPLE 1 Find the length, midpoint and gradient of AB for $A(-2, 3)$ and $B(4, -5)$.

$$\textcircled{1} \quad \Delta x = 6, \Delta y = -8$$

$$\textcircled{2} \quad AB = \sqrt{36 + 64} = 10$$

$$\textcircled{3} \quad M = (1, -1)$$

$$\textcircled{4} \quad m = \frac{-8}{6} = -\frac{4}{3}$$

ANSWER

$$10, (1, -1), -\frac{4}{3}$$

EXAMPLE 2 Find the equation of the line through $(2, -1)$ perpendicular to $y = \frac{3}{4}x + 2$.

$$\textcircled{1} \quad m_1 = \frac{3}{4} \Rightarrow m_2 = -\frac{4}{3}$$

Negative reciprocal.

$$\textcircled{2} \quad y + 1 = -\frac{4}{3}(x - 2)$$

$$\textcircled{3} \quad 3y + 3 = -4x + 8$$

$$\textcircled{4} \quad 4x + 3y - 5 = 0$$

ANSWER

$$4x + 3y - 5 = 0$$

EXAMPLE 3 Find the perpendicular bisector of $A(1, 2)$ and $B(5, 8)$.

$$\textcircled{1} \quad M = (3, 5)$$

$$\textcircled{2} \quad m_{\{AB\}} = \frac{6}{4} = \frac{3}{2}$$

$$\textcircled{3} \quad m = -\frac{2}{3}$$

$$\textcircled{4} \quad 2x + 3y = 21$$

ANSWER

$$2x + 3y = 21$$

EXAMPLE 4 Show that $A(1, 1)$, $B(3, 5)$ and $C(6, 11)$ are collinear.

① $m_{AB} = \frac{4}{2} = 2$

② $m_{BC} = \frac{6}{3} = 2$

- ③ Equal gradients and a common point B .
So one line.

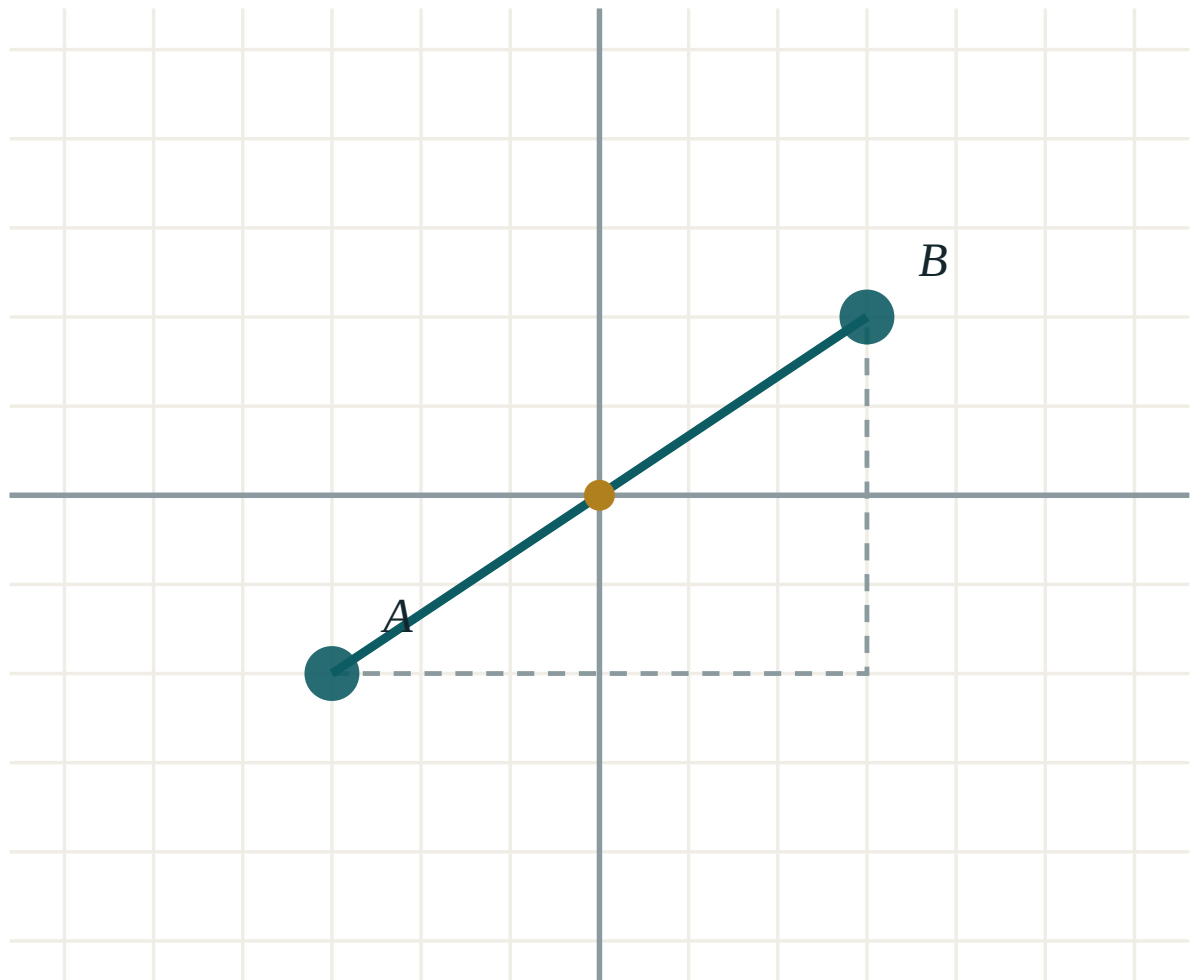
ANSWER

Collinear — both gradients are 2

03 Interactive model

Plot the three points of the collinearity example on squared paper before computing anything.

Gradient, midpoint and length



A (-3, -2)

B (3, 2)

 $x_2 - x_1$ 6 $y_2 - y_1$ 4

distance AB 7.21

midpoint M (0, 0)

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Coordinate plane	Koordinata tekisligi	Координатная плоскость
Gradient (slope)	Burchak koeffitsiyenti	Угловой коэффициент
Midpoint	O'rta nuqta	Середина
Distance formula	Masofa formulasi	Формула расстояния
Parallel lines	Parallel to'g'ri chiziqlar	Параллельные прямые
Perpendicular lines	Perpendikulyar to'g'ri chiziqlar	Перпендикулярные прямые
Perpendicular bisector	O'rta perpendikulyar	Серединный перпендикуляр
Intercept	Kesma	Отрезок на оси
Collinear	Bir to'g'ri chiziqda	Коллинеарные
Foot of the perpendicular	Perpendikulyar asosi	Основание перпендикуляра

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The gradient of the line through $(1, 2)$ and $(4, 8)$ is:

Perpendicular gradients satisfy:

The line $x = 3$ has gradient:

The midpoint of $(-2, 3)$ and $(4, -5)$ is:

$(1, -1)$

$(3, -4)$

$(2, -2)$

$(1, 4)$

A line parallel to $3x + 2y = 7$ is:

$2x - 3y = 1$

$3x + 2y = 13$

$3x - 2y = 7$

$2x + 3y = 7$

The perpendicular bisector passes through:

A

B

the midpoint

the origin

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Length from $(0, 0)$ to $(3, 4)$

ANSWER 5

2. Midpoint of $(1, 2)$ and $(5, 8)$

ANSWER $(3, 5)$

3. Gradient through $(1, 2)$ and $(4, 8)$

ANSWER 2

4. Gradient of $y = -3x + 1$

ANSWER -3

5. Gradient perpendicular to 2

ANSWER $-\frac{1}{2}$

6. Equation through $(0, 4)$ with gradient 3

ANSWER $y = 3x + 4$

7. Gradient of $y = 5$

ANSWER 0

Medium · the standard question

7 PROBLEMS

1. Length from $(-2, 3)$ to $(4, -5)$

ANSWER 10

2. Equation through $(2, -1)$ with gradient $\frac{3}{4}$

ANSWER $3x - 4y - 10 = 0$

3. Equation through $(2, -1)$ perpendicular to $y = \frac{3}{4}x$

ANSWER $4x + 3y - 5 = 0$

4. Line through $(1, 3)$ and $(5, 11)$

ANSWER $y = 2x + 1$

5. Perpendicular bisector of $(1, 2)$ and $(5, 8)$

ANSWER $2x + 3y = 21$

6. Are $(1,1)$, $(3,5)$, $(6,11)$ collinear?

ANSWER Yes, gradient 2

7. Line parallel to $3x + 2y = 7$ through $(1, 5)$

ANSWER $3x + 2y = 13$

Hard · stretch and reason

7 PROBLEMS

1. Foot of the perpendicular from $(1, 8)$ to $y = 2x$

ANSWER $(3.4, 6.8)$

2. Distance from $(1, 8)$ to the line $y = 2x$

ANSWER $\frac{6}{\sqrt{5}} \approx 2.68$

3. Area of the triangle with vertices $(0,0)$, $(4,0)$, $(2,6)$

ANSWER 12

4. Find k so that $(1, 2)$, $(3, k)$, $(7, 14)$ are collinear

ANSWER $k = 6$

5. The vertices $A(0,0)$, $B(4,2)$, $C(2,6)$: show the triangle is right-angled

ANSWER $m_{AB} m_{BC} = \frac{1}{2} \times (-2) = -1$

6. Circumcentre of $(0,0)$, $(6,0)$, $(0,8)$

ANSWER $(3, 4)$

7. The line $y = mx$ is equidistant from $(0, 4)$ and $(6, 0)$: find m

ANSWER $m = \frac{2}{3}$

07 Homework

Homework — 5 tasks

Sketch the points before every calculation; a wrong sign shows up in the picture at once.

- Find the length, midpoint and gradient of AB for $A(-3, 1)$ and $B(5, 7)$.
- Find the equation of the line through $(3, -2)$ parallel to $2x - 5y = 4$.
- Find the equation of the perpendicular bisector of $(-1, 4)$ and $(5, -2)$.
- Show that $A(1, 0)$, $B(4, 6)$ and $C(6, 10)$ are collinear.

5. Find the foot of the perpendicular from $(2, 9)$ to the line $y = 3x + 1$.